PYTHON · FROM ZERO · LEARNING LOG

Python Study Pack

All my lessons, revision guides and drills in one place. **Start with Revision #1**, and cover the answers before checking them.

7	26 Oct 2026
guides inside	last updated

📚 Contents

🧭 **Revision #1: The Journey So Far (Days 01–07 + 23 problems)**
Start here: the 9 revision topics, checklists, what to rewrite

📕 **The Add · Remove · Loop Book**
Drills: position vs value, add / remove / loop

🧠 **Data Structures Guide: Lists, Strings, Dictionaries & Parsing Input**
What am I holding? Nested data, parsing, printing

🧘 **How to Solve a Problem Without Panicking**
The STEP method: solve without panicking

🔁 **Big Review #1: Sat 24 + Sun 25 Oct 2026**
Predict-the-output drills: strings, map, dicts

📘 **Day 06: Functions 🧩**
Day 06 lesson: functions

📘 **Day 07: Dictionaries 📖**
Day 07 lesson: dictionaries

🧭 Revision #1: The Journey So Far (Days 01–07 + 23 problems)

↑ contents

> **Goal:** before Day 08, make everything from the last 2 weeks feel solid. Then a **long quiz: 10–15 questions per topic**, one topic at a time. Everything here comes from **my** lessons, problems and mistakes.

🗺️ The journey in one table

Week	Lessons	Problems	Big moments
1 (12–18 Oct)	Days 01–06, Day 07 parts 1–2	9 HackerRank, 2 LeetCode	⭐⭐ 2nd star, first LeetCode, STEP-comments method
2 (19–25 Oct)	Day 07 parts 3–4 + Two Sum bonus	3 HackerRank, 7 LeetCode	first Mediums (2149, Word Order), HackerRank Orchestrate hackathon
3 (26 Oct–)	revision	LeetCode 1689, Piling Up!	⭐⭐⭐⭐ 4th star, LeetCode Day 10 🔥

📋 How to revise (per topic)

1. 📖 **Read** the listed sections (read, don't copy). Tick each ✅ line in **"I can..."** only if you could do it with no notes.
2. 🐞 **Reread your own mistakes** for that topic. These are the questions most likely to come up in the quiz 😄
3. ✍️ **Rewrite from memory** the listed problem in a scratch file (`review.py`). Stuck for 5 min? Peek at **one line** only.
4. 💬 Say **"ready for topic N"** and you'll get the quiz: 10–15 questions mixing *predict the output*, *find the bug*, *write a few lines* and *explain it like a YouTube viewer*.

Scoring: ≥ 80% = ✅ topic done · 60–79% = retry the ❌ ones next session · < 60% = reread and redo that topic. Every quiz is logged in quiz-log.md.

Suggested order (3–4 sessions, at whatever pace life allows): - **Session A:** Topics 1–3 (basics, conditions, loops) - **Session B:** Topics 4–5 (strings, lists) ⭐ - **Session C:** Topics 6–8 (functions, dictionaries, parsing and printing) ⭐ - **Session D:** Topic 9 (patterns, errors, problem solving), then **Day 08** 🎉

Warm-up drills you already have: review-weekend-1.md (strings, `map`, dicts) and add-remove-loop-book.md (add, remove, loop).

1️⃣ Basics: print, variables, input, types, maths (Days 01–02)

📖 **Read:** notes.md: *Day 01*, *Day 02*, *Maths operators* · day01.py, day02.py **I can...** - ☐ explain what a variable is and print several values with commas - ☐ say what type `input()` gives, and convert with `int()`, `float()`, `str()` - ☐ write an f-string with values inside `{}`, and control spaces exactly - ☐ predict `7 / 2`, `7 // 2`, `7 % 2`, `2 ** 3`, and use `% 2 == 0` for "even" - ☐ explain why `"12" + 3` crashes and fix it two ways

🐞 **My mistakes:** missing comma in `print` · f-string spacing `is{age}` · commas in `print` aren't printed ✍️ **Rewrite:** HackerRank *Arithmetic Operators* and *Python: Division*

2️⃣ Conditions: if / elif / else (Day 02)

📖 **Read:** notes.md: *Day 02* · approaches.md: *Python If-Else* **I can...** - ☐ write a flat `if / elif / else` chain (no nesting) and explain why `elif` can **assume** the earlier checks failed - ☐ use `and`, `or`, `not`, `in`, and chained ranges `2 ≤ n ≤ 5` - ☐ pick `>` vs `≥` from the wording ("90+" means `≥ 90`) - ☐ explain why `ch == "a" or "e"` is always True and fix it

🐞 **My mistakes:** `|` used as "or" · `grade == "A"` instead of `"A"` · boundary 90 excluded ✍️ **Rewrite:** HackerRank *Python If-Else* · the Day 06 `grade()` function

3️⃣ Loops: for, range, while, break, continue (Day 03)

📖 **Read:** notes.md: *Day 03*, *Loop keywords* · day03.py **I can...** - ☐ list exactly what `range(5)`, `range(1, 6)`, `range(2, 21, 2)`, `range(5, 0, -1)` give - ☐ choose `for` (known count) vs `while` (until something changes) - ☐ make sure every `while` loop changes something so it can end - ☐ explain `break` vs `continue` vs `pass` - ☐ build a total with an ACCUMULATOR (`total = 0`, `total += x`) - ☐ write a loop that goes **backwards** over positions

🐞 **My mistakes:** `range(0, n+1)` off-by-one (HackerRank Loops) · infinite loop (344) · `range(1:n+1)` with a colon (412) · missing `break` (Runner-Up) ✍️ **Rewrite:** HackerRank *Loops* · sum 1 to 100 (three ways: loop, `sum`, Gauss)

4️⃣ Strings ⭐ (Day 04)

📖 **Read:** notes.md: *Day 04* · data-structures-guide.md §2, §4, §6 · add-remove-loop-book.md §4 · review-weekend-1.md *Saturday* drills **I can...** - ☐ index and slice: `s[0]`, `s[-1]`, `s[1:4]`, `s[::-1]`, `s[1:]`, `s[:-1]` - ☐ explain why strings can't be changed (`s[0] = "X"` crashes) and rebuild them with `+=` - ☐ use `.upper()` `.lower()` `.strip()` `.replace()` `.split()` `" ".join()`, and remember they **return a new string** - ☐ search 3 ways: `x in s` (yes/no), `s.find(x)` (position or -1), `s.count(x)` (how many) - ☐ choose a **letters loop** vs a **positions loop** - ☐ compare strings case-insensitively

🐞 **My mistakes:** `word[ch]` when `ch` is a letter · `.index[i]` with square brackets · `Racecar` failing without `.lower()` · `string.find(ch) == 0` (Merge the Tools) ✍️ **Rewrite:** palindrome · vowel counter · LeetCode 1108 *Defanging an IP* · HackerRank *Merge the Tools!*

5️⃣ Lists ⭐⭐ (Day 05)

📖 **Read:** notes.md: *Day 05* · add-remove-loop-book.md §0–§2 (**all of it**) · data-structures-guide.md §3 **I can...** - ☐ answer "**position or value?**" for `pop`, `insert`, `remove`, `del`, `in`, `index` - ☐ grow a list with `.append()` (never `ans[0] = ...` on an empty list) - ☐ say what each method **gives back** (`pop` → the item; `append`, `sort`, `remove` → `None`) - ☐ explain `sorted(x)` vs `x.sort()`, and functions (around) vs methods (after the dot) - ☐ loop 4 ways: items, `range(len( ... ))`, `enumerate`, two pointers - ☐ read nested lists: `students[0][1]` - ☐ explain why removing from a list while looping over it skips items

🐞 **My mistakes (9 items-vs-positions slips!):** even filter · 1929 · 1480 `output[value - 1]` · 2149 one counter for two lists · `range(list)` ×3 · `scores.max()` · `pop(blocks[0])` (Piling Up) ✍️ **Rewrite:** second-highest score (with duplicates) · LeetCode 1929, 2149 · HackerRank *Lists*, *Piling Up!*

6️⃣ Functions (Day 06)

📖 **Read:** notes.md: *Day 06* · lessons/day06-functions.md · approaches.md: *Day 06 functions* **I can...** - ☐ write `def name(params):` with a default value, and call it - ☐ explain `return` vs `print`, and why a function with no `return` gives `None` - ☐ explain `break` vs `return` (loop vs whole function) - ☐ spot a **side effect** (`scores.sort()` inside a function changes the caller's list) - ☐ say why a variable must never share a function's name (`grade = ...`) - ☐ write a GUARD CLAUSE for empty input

🐞 **My mistakes:** `print` instead of `return` (×3) · `return {x}` makes a set · dead code after `return` · `return` outside a function (Nested Lists, Piling Up) ✍️ **Rewrite:** `count_vowels`, `is_palindrome`, `average`, `second_highest`, `grade` as functions that **return**

7️⃣ Dictionaries ⭐ (Day 07)

📖 **Read:** notes.md: *Day 07* · lessons/day07-dictionaries.md · add-remove-loop-book.md §3 · review-weekend-1.md *Sunday* drills **I can...** - ☐ add, change, look up and delete: `d[k] = v`, `d[k]`, `d.get(k, 0)`, `del d[k]`, `d.pop(k)` - ☐ say what `key in d` checks (**keys only**) - ☐ loop over keys, values and both (`.items()`) - ☐ write the COUNTING line from memory: `counts[x] = counts.get(x, 0) + 1` - ☐ find the most common item with BEST SO FAR (updating **both** variables) - ☐ explain why `d["z"] += 1` crashes on a new key

🐞 **My mistakes:** `best_count` never updated (letter counter) · counted letters instead of words (Word Order) · `counts.items` without `()` ✍️ **Rewrite:** letter counter + most common letter · Two Sum with a `seen` dict · HackerRank *Word Order*

8️⃣ Parsing input and printing ⭐ (HackerRank's favourite)

📖 **Read:** data-structures-guide.md §5–§6 · add-remove-loop-book.md §5 · notes.md: *Reading many numbers from one line* **I can...** - ☐ read one number, a line of numbers, and "a name then numbers" (`name, *rest = input().split()`) - ☐ explain `list(map(int, input().split()))` piece by piece, and why `map` needs `list( ... )` - ☐ read T test cases with a loop - ☐ print on one line 3 ways: `end=" "`, `" ".join(map(str, nums))`, `print(*nums)` - ☐ use `sep=` and f-string formatting `f"{avg:.2f}"`

🐞 **My mistakes:** `map( ... ).sort()` · `input.split` without `()` · used the loop's last `name` instead of `query_name` (Finding the percentage) ✍️ **Rewrite:** HackerRank *Find the Runner-Up Score!*, *Finding the percentage*, *Nested Lists*

9️⃣ Patterns, errors and problem solving

📖 **Read:** approaches.md: *Pattern library* + any 3 problem sections · lessons/problem-solving-method.md · mistakes-journal.md: *Focus areas* + *Daily checklist* · cheat sheet §10–§12 **I can...** - ☐ name and write each pattern from memory: ACCUMULATOR, FILTER, COUNTING, BEST SO FAR, TWO POINTERS, EARLY EXIT, GUARD CLAUSE, FLAG, TAKE FROM BOTH ENDS, FIND THE BOTTLENECK - ☐ look at a new problem and say which pattern fits - ☐ use the STEP method: solve by hand → steps in English → code one step at a time - ☐ decode errors: `SyntaxError` (missing `:`, `return` outside a function), `IndexError`, `KeyError`, `TypeError` (`str + int`, `range(list)`, `word[ch]`), `AttributeError` (`map.sort`, `scores.max`), "Time Limit Exceeded" (infinite loop) - ☐ explain the LeetCode format (`class Solution`, **return**, inputs already typed) vs HackerRank (read `input()`, **print**) - ☐ use the contest routine: read twice → example by hand → STEP comments → code → **Run** → Submit

✍️ **Rewrite:** LeetCode 344 (two pointers) · 1480 (accumulator) · 1689 (bottleneck) · 2161 (filter)

🏁 After the quiz

- Every ❌ goes to quiz-log.md's retry list and gets a drill in add-remove-loop-book.md if it's about lists, dicts or strings.
- All topics ✅ → **Day 08: sets and tuples** 🎉
- Daily minimum during revision: **1 LeetCode** (streak) and **1 commit**. The rainy-day bank in progress.md helps on busy days.

📕 The Add · Remove · Loop Book

↑ contents

> My weak spot, attacked one chapter at a time: **putting things into, taking things out of, and walking through** lists, dictionaries and strings. **How to use it:** 3 drills from the 🏋️ sections at the start of every session (answers are at the bottom, so cover them up!). A new chapter gets added whenever a problem trips me up. Companion to data-structures-guide.md, which covers parsing input and printing.

---

0. The one question that fixes most bugs

Before touching a list, ask: **"Do I mean a POSITION or a VALUE?"**

```
cubes  =  [ 4,  3,  2,  1 ]
position:   0   1   2   3      ← where it sits (cubes[0])
value:      4   3   2   1      ← what it is
```

Wants a position	Wants a value
`lst[i]`	`x in lst`
`lst.pop(i)`	`lst.remove(x)`
`lst.insert(i, x)` (i = where)	`lst.append(x)` (x = what)
`del lst[i]`	`lst.count(x)`
`range(len(lst))` gives positions	`lst.index(x)` gives back a position

🔁 **My repeat offenders:** `pop(blocks[0])` (Piling Up), `output[value - 1]` (1480), `range(list)` (1672, Piling Up).

---

1. 📋 Lists

Add

Code	What happens	Gives back
`lst.append(x)`	puts x at the **end**	`None`
`lst.insert(i, x)`	puts x **at position i**, pushes the rest right	`None`
`lst + [x, y]`	makes a **new** list, glued together	the new list
`lst.extend([x, y])`	adds each item to the end	`None`

Remove

Code	What happens	Gives back
`lst.pop()`	removes the **last** item	**that item** ✅
`lst.pop(0)`	removes the **first** item	**that item** ✅
`lst.pop(i)`	removes the item **at position i**	**that item** ✅
`lst.remove(x)`	removes the **first item equal to x**	`None`
`del lst[i]`	removes the item at position i	nothing (it's a statement)

Look / change

Code	Meaning
`lst[0]`, `lst[-1]`	first, last
`lst[len(lst) - 1]`	last, the long way (same as `lst[-1]`)
`lst[i] = x`	replace what's at position i
`lst[a:b]`	a **copy** of positions a … b-1

⚠️ Traps

1. **`lst = lst.append(x)`** → lst becomes `None`! `append` changes the list *in place* and gives back nothing. Just write `lst.append(x)`.
2. **`print(lst.sort())`** prints `None` for the same reason. Use `print(sorted(lst))`, or `lst.sort()` and then `print(lst)`.
3. **Removing while looping over the same list** skips items. Loop with `while lst:` instead, or build a new list (FILTER pattern).
4. **`pop()` on an empty list** → `IndexError: pop from empty list`. Check `if lst:` first.

🏋️ Drills: lists (predict the output, then run it)

```
# L1
a = [10, 20, 30]
a.append(40)
print(a)

# L2
a = [10, 20, 30]
x = a.pop(0)
print(x, a)

# L3
a = [10, 20, 30]
a.insert(1, 99)
print(a)

# L4
a = [5, 3, 5, 1]
a.remove(5)
print(a)

# L5
a = [4, 3, 2]
a = a.append(1)
print(a)

# L6
a = [4, 3, 2, 1]
print(a.pop(a[0]))          # careful!
```

2. 🔁 Walking through a list: 4 ways

I need...	Write	Each round gives me
just the items	`for x in lst:`	the value
positions (to look left/right, or change `lst[i]`)	`for i in range(len(lst)):`	the position
both	`for i, x in enumerate(lst):`	position **and** value
to eat it from the ends	`while lst:` + `lst.pop()` / `lst.pop(0)`	whatever you pop
two ends moving inwards	`left, right = 0, len(lst) - 1` + `while left ≤ right:`	two positions

🏋️ Drills: loops

```python
# W1: what's printed?
for i in range(len(["a", "b", "c"])):
    print(i)

# W2: what's printed?
for i, x in enumerate(["a", "b"]):
    print(i, x)

# W3: what's printed?
s = [1, 2, 3]
while s:
    print(s.pop())

# W4: fix the bug
nums = [3, 1, 2]
for i in range(nums):
    print(nums[i])
```

---

3. 📖 Dictionaries

A dictionary maps a **key** → **value**. There are no positions, **only keys**.

Add / change / remove / look

Code	What happens
`d[key] = value`	adds the key, or **overwrites** it if it exists
`d[key] += 1`	⚠️ crashes with `KeyError` if the key isn't there yet
`d[key] = d.get(key, 0) + 1`	the **COUNTING** line: safe even for a new key
`d[key]`	look up (⚠️ `KeyError` if missing)
`d.get(key, default)`	look up, with no crash
`key in d`	checks **keys**, not values
`d.pop(key)`	removes the key and **gives back its value**
`del d[key]`	removes the key

Walking through

I need...	Write
keys	`for k in d:`
values	`for v in d.values():`
both	`for k, v in d.items():`
keys in order	`for k in sorted(d):`

🏋️ Drills: dictionaries

```
# D1
d = {}
d["a"] = 1
d["a"] = 5
print(d)

# D2
d = {"x": 1}
print("x" in d, 1 in d)

# D3
counts = {}
for ch in "banana":
    counts[ch] = counts.get(ch, 0) + 1
print(counts)

# D4
d = {"a": 1, "b": 2}
v = d.pop("a")
print(v, d)

# D5: what happens?
d = {}
d["z"] += 1
```

---

4. 🔤 Strings: you can't change them, only rebuild

Strings are **frozen** (*immutable*): `s[0] = "X"` crashes. To "add" or "remove", you build a **new** string.

Want	Write
add to the end	`s = s + "x"` or `s += "x"`
remove the first char	`s = s[1:]`
remove the last char	`s = s[:-1]`
list of words	`s.split()`
words → one string	`" ".join(words)`
list of characters	`list(s)`

🏋️ Drills: strings

```
# S1
s = "cat"
s += "s"
print(s)

# S2
print("hello"[1:-1])

# S3
print("-".join(["a", "b", "c"]))
```

---

5. 🗺️ `map`: do the same thing to every item

`map(f, items)` = "apply **f** to each item". It gives back a lazy *map object*, so wrap it in `list( ... )` to see it.

```
words = "4 3 2".split()          # ['4', '3', '2']   ← strings!
nums  = list(map(int, words))    # [4, 3, 2]          ← numbers
```

It's the same as this loop:

```
nums = []
for w in words:
    nums.append(int(w))
```

🏋️ Drills: map

```
# M1
print(list(map(int, "7 8 9".split())))

# M2
print(list(map(len, ["hi", "hello"])))

# M3: what's printed (and why is it odd)?
print(map(int, ["1", "2"]))
```

---

6. 📒 Chapters added from real problems

Problem	What tripped me	Chapter
HackerRank Piling Up! (26 Oct)	`pop(value)` instead of `pop(position)`; `range(list)`	§0, §1
LeetCode 1672 (24 Oct)	`range(list)`	§2
HackerRank Merge the Tools (23 Oct)	rebuilding a string per piece	§4
LeetCode 2149 (22 Oct)	mixing positions of two lists of different sizes	§0
HackerRank Word Order (22 Oct)	counting letters instead of words	§3

✅ Answers (cover these up!)

Click to reveal

- **L1** `[10, 20, 30, 40]`
- **L2** `10 [20, 30]`

- **L3** `[10, 99, 20, 30]`
- **L4** `[3, 5, 1]` : only the **first** 5 goes
- **L5** `None` : `append` gives back nothing
- **L6** `IndexError` : `a[0]` is 4, so it tries `pop(4)` , but the last position is 3
- **W1** `0 1 2` (one per line)
- **W2** `0 a` then `1 b`
- **W3** `3 2 1` : pop takes from the end
- **W4** `range(len(nums))`
- **D1** `{'a': 5}` : same key, overwritten
- **D2** `True False` : `in` checks keys only
- **D3** `{'b': 1, 'a': 3, 'n': 2}`
- **D4** `1 {'b': 2}`
- **D5** `KeyError: 'z'` : use `d.get("z", 0) + 1`
- **S1** `cats`
- **S2** `ell`
- **S3** `a-b-c`
- **M1** `[7, 8, 9]`
- **M2** `[2, 5]`
- **M3** `<map object at 0x ... >` : needs `list( ... )` around it

📦 Data Structures Guide: Lists, Strings, Dictionaries & Parsing Input

↑ contents

> ⭐ **My weak spot.** Read this before every HackerRank / LeetCode problem until it feels natural. Golden rule: **always know WHAT you're holding.** Is it a string? A list? A dictionary? A list *inside* a dictionary? 📕 **Adding, removing and looping** (append/pop/insert/remove, dict add/remove, `map`) now has its own drill book: add-remove-loop-book.md.

---

1. The 3 containers at a glance

	String `" ... "`	List `[ ... ]`	Dictionary `{ ... }`
Holds	characters	any items, in order	**key** → **value** pairs
Example	`"Harsh"`	`[25.0, 26.5, 28.0]`	`{"Harsh": [25.0, 26.5, 28.0]}`
Get one item	`s[0]` → `"H"`	`x[0]` → `25.0`	`d["Harsh"]` → `[25.0, 26.5, 28.0]`
By	position	position	**key** (name)
Can change?	❌ no	✅ yes	✅ yes
Length	`len(s)`	`len(x)`	`len(d)` (number of pairs)
Check inside	`"H" in s`	`25.0 in x`	`"Harsh" in d` (checks **keys**)

---

2. 🔍 "What am I holding?" Print it!

When confused, **print the variable and its type**. This is the #1 debugging trick:

```
print(student_marks)          # {'Harsh': [25.0, 26.5, 28.0], 'Anurag': [26.0, 28.0, 30.0]}
print(type(student_marks))    # <class 'dict'>
print(type(student_marks["Harsh"]))     # <class 'list'>
print(type(student_marks["Harsh"][0]))  # <class 'float'>
```

---

3. 🪜 Going down layers (nested data)

Read nested access **left to right, one step at a time**:

```
student_marks = {"Harsh": [25.0, 26.5, 28.0], "Anurag": [26.0, 28.0, 30.0]}

student_marks                 # the whole dictionary
student_marks["Harsh"]        # step 1: open the "Harsh" drawer → [25.0, 26.5, 28.0]
student_marks["Harsh"][1]     # step 2: second item in that list → 26.5
```

Like an address: **cupboard** → **drawer** → **item**.

---

4. 🔁 Looping over each container

Loop	Gives you	Example
`for ch in "hi":`	each **character**	`h`, `i`
`for x in [10, 20]:`	each **item**	`10`, `20`
`for i in range(len(lst)):`	each **position**	`0`, `1`, then use `lst[i]`
`for key in d:`	each **key**	`"Harsh"`, `"Anurag"`
`for value in d.values():`	each **value**	`[25.0, ...]`, `[26.0, ...]`
`for key, value in d.items():`	**both** ⭐	`"Harsh"`, `[25.0, ...]`

🔑 **Name the loop variable after what it IS:** `ch`, `num`, `name`, `marks`. Use `i` only for positions.

⚠️ **The loop variable keeps its LAST value after the loop ends:**

```
for name in ["Harsh", "Anurag"]:
    pass
print(name)       # "Anurag", the last one! (this caused my "Finding the percentage" bug)
```

---

5. 📥 Parsing input (HackerRank's favourite)

Input always arrives as **text**. Turning it into numbers and containers is called **parsing**.

One number

```
n = int(input())                    # "3" → 3
```

Many numbers on one line

```
nums = list(map(int, input().split()))
```

Read it **inside** → **out** with input `"2 3 6"`: | Step | Code | Result | |---|---|---| | 1 | `input()` | `"2 3 6"` (one string) | | 2 | `.split()` | `["2", "3", "6"]` (list of strings) | | 3 | `map(int, ...)` | applies `int()` to each (a lazy "recipe") | | 4 | `list(...)` | `[2, 3, 6]` (real list of numbers) ✅ |

Use `float` instead of `int` for decimals: `list(map(float, ...))` → `[25.0, 26.5]`. Same thing without `map` (a list comprehension, coming soon): `[int(x) for x in input().split()]`

A name followed by numbers

```
name, *rest = "Harsh 25 26.5 28".split()
# name = "Harsh"
# rest = ["25", "26.5", "28"]     (* = "collect everything else into a list")
scores = list(map(float, rest))   # [25.0, 26.5, 28.0]
```

Many lines → a dictionary

```
student_marks = {}
for _ in range(n):                  # _ = "I don't need the counter, just repeat n times"
    name, *line = input().split()
    student_marks[name] = list(map(float, line))
```

---

6. 🖨️ Printing

```python
x = [1, 2, 3]
print(x)                    # [1, 2, 3]       (the whole list, with brackets)
print(*x)                   # 1 2 3           (* "unpacks" the items, separated by spaces)
print(*x, sep=",")          # 1,2,3
for item in x:
    print(item)             # one per line

d = {"a": 1, "b": 2}
for key, value in d.items():
    print(f"{key}: {value}") # a: 1   b: 2

avg = 26.5
print(f"{avg:.2f}")         # 26.50 (2 decimal places)
```

---

7. 🧰 The patterns that cover 80% of problems

Pattern	Container	Code shape
ACCUMULATOR	number	`total = 0` → `total += x`
BUILD A LIST	list	`result = []` → `result.append(x)`
FILTER	list	`if condition: result.append(x)`
COUNTING	dict	`counts[x] = counts.get(x, 0) + 1`
LOOKUP	dict	`d[key]` or `d.get(key, default)`
BEST SO FAR	any	`best = None` → `if best is None or x > best: best = x`

---

8. ✅ Before submitting: data checklist

- ☐ Did I **print** the data once to see its real shape?
- ☐ Am I using the **right variable**? (e.g. `query_name`, not a leftover loop variable)
- ☐ Is it **text or a number**? Do I need `int()` / `float()`?
- ☐ Am I getting **items, positions or keys** from this loop?
- ☐ Am I using the data **after** it's been read? (order of lines matters!)

🧘 How to Solve a Problem Without Panicking

↑ contents

> "I can solve the logic, but I can't write the code" is **normal**. Logic and code are two different skills. This method connects them. **Never write the whole program at once.**

The 6-step method

1. Solve the sample by hand 🖍️

Read the sample input and work out the answer on paper **without code**. If you can't do it by hand, you can't code it yet.

2. Write the steps as comments 📝

Turn your hand-solution into short English steps, as `#` comments in the editor:

```
# STEP 1: read all students into a list of [name, score]
# STEP 2: make a sorted list of just the scores
# STEP 3: find the second-lowest score
# STEP 4: collect the names with that score
# STEP 5: sort the names and print each one
```

Now the big scary problem is **5 small problems**.

3. For each step, ask "what am I holding?" 🔍

Write the **shape** of the data next to each step:

```
# students = [["Harry", 37.21], ["Tina", 37.2], ...]   (list of lists)
# scores   = [37.2, 37.21, 37.21, 39.0, 41.0]          (list of numbers)
# second   = 37.21                                       (one number)
# names    = ["Berry", "Harry"]                          (list of strings)
```

4. Code ONE step, then `print()` and run ▶️

Write only Step 1, add `print(students)`, click **Run**. Does the output match the shape you wrote? ✅ Then move on. ❌ Fix just this step. **Small steps = small errors = no panic.**

5. Translate English → Python with the "phrasebook" 📖

When I want to...	Python
start an empty list	`x = []`
add something to a list	`x.append(thing)`
go through every item	`for item in x:`
keep only some items	`if condition:` inside the loop, then `append`
sort a list	`x.sort()` or `sorted(x)`
get the first / last	`x[0]` / `x[-1]`
inside a pair `[name, score]`	`pair[0]` = name, `pair[1]` = score
stop at the first match	`break`
count things	`counts[k] = counts.get(k, 0) + 1`
print each item on its own line	`for item in x: print(item)`

6. Clean up and submit 🧹

Remove the debug `print()`s, test the sample once more, and submit. If it fails, look at the failing input and ask: "what's different about it?"

🚑 When panic hits

- Stop typing. Breathe. **Go back to step 1** (by hand).
- Delete nothing. Comment the broken code out with `#` and start the current step again.
- A red squiggle = **one** small problem. Fix the **first** one only, then re-run.
- `return` only works inside `def`. On HackerRank's main code, use `print`.

Common syntax traps from my journal

- Filling an empty list with `x[0] = ...` → use `x.append(...)`
- `x.sorted()` ❌ → `sorted(x)` or `x.sort()`
- `for item in range(len(x))` gives **positions**, so use `x[item]` to get the value
- `return` outside a function ❌ → `print()`

🔁 Big Review #1: Sat 24 + Sun 25 Oct 2026

↑ contents

> ℹ️ Not done on the weekend (hackathon!). These drills are now **warm-ups for revision-1.md**, topics 4, 5, 7 and 8. Goal: make **strings, searching, printing, items vs positions, `map` and dictionaries** feel automatic. These are my own weak spots (said on 23 Oct). Everything here comes from **my** lessons, problems and mistakes.

How each block works (about 1.5–2 h per day, take breaks): 1. 📖 **Reread** (10 min): the listed notes. Read, don't copy. 2. 🧠 **Predict** (20 min): for each drill, write what it prints **before** running it. Then check in VS Code (`review.py`, a scratch file). 3. ✍️ **Rewrite from memory** (40 min): solve old problems again **without looking**. Stuck for 5 min? Peek at one line only. 4. 💡 **Better ways** (15 min): read the matching sections in approaches.md. 5. 🟢 **One Easy LeetCode** to keep the streak. Then commit.

Mark each drill ✅ (right first time), 🟡 (right after thinking) or ❌ (wrong). The ❌ ones go to quiz-log.md's retry list.

🗓️ Saturday: Strings, searching, printing, items vs positions

📖 Reread

- notes.md: Day 03 (loops), Day 04 (strings), "print tricks"
- data-structures-guide.md sections 2, 4 and 6
- mistakes-journal.md: focus area 1 (items vs positions)

🧠 Predict the output

```
s = "banana"
# S1
print(s[1], s[-1], s[1:4])
# S2
print(s.find("n"), s.find("z"))
# S3
print(s.count("a"), "nan" in s, "N" in s)
# S4
s.upper()
print(s)
# S5
print("a-b-c".split("-"))
# S6
print("-".join(["x", "y", "z"]))
# S7
for ch in "abc":
    print(ch, end=" ")
# S8
for i in range(len("abc")):
    print(i, end=" ")
# S9
nums = [5, 7, 9]
for i in range(len(nums)):
    print(i, nums[i])
# S10
print(1, 2, 3, sep=", ")
# S11
print("A", end="")
print("B")
# S12
result = ""
for ch in "hello":
    if ch not in result:
        result += ch
print(result)
```

S13 (choose the loop): you need to print the **position** of every `"a"` in `"banana"`. Items loop or positions loop? Why? Write it.

S14 (find the bug):

```
word = "level"
for ch in word:
    if word[ch] == "e":
        print("found")
```

✍️ Rewrite from memory

1. Day 04: **vowel counter** (count vowels in a word, any case)
2. Day 04: **palindrome check**
3. LeetCode 344: **Reverse String** in place (two pointers)
4. HackerRank: **Merge the Tools!** (try it with no peeking now)

💡 Better ways

approaches.md: Palindrome, Vowel counter, 344, Merge the Tools

---

🗓️ Sunday: `map`, parsing input, dictionaries (+ 🗓️ Sunday review)

📖 Reread

- notes.md: "Reading many numbers from one line", Day 07 (dictionaries)
- data-structures-guide.md sections 1, 3 and 5
- lessons/day07-dictionaries.md: the tally-sheet part

🧠 Predict the output

```
# M1  (pretend you typed: 4 5 6)
line = "4 5 6"
print(line.split())
# M2
print(map(int, line.split()))
# M3
print(list(map(int, line.split())))
# M4
print(list(map(str, [1, 2, 3])))
# M5
print(" ".join(map(str, [1, 2, 3])))
# M6
name, *rest = "Harsh 25 26".split()
print(name, rest)

d = {"a": 1, "b": 2}
# D1
print(d["a"], d.get("z"), d.get("z", 0))
# D2
for k in d:
    print(k, end=" ")
# D3
for k, v in d.items():
    print(k, v)
# D4
print(list(d.values()), len(d))
# D5
print("a" in d, 1 in d)
# D6
d["c"] = 3
d["a"] = 10
print(d)
# D7
counts = {}
for ch in "hello":
    counts[ch] = counts.get(ch, 0) + 1
print(counts)
```

D8 (what happens?): `print(d["z"])`

D9 (find the bug):

```
counts = {}
for word in ["hi", "yo", "hi"]:
    counts[word] += 1
```

D10 (choose): you have `nums = [3, 8, 3]` and must print `3 8 3` on one line. Write it **two** ways.

✍️ Rewrite from memory

1. Day 07 Part 3: **letter counter + most common letter**
2. HackerRank: **Word Order**
3. HackerRank: **Finding the percentage** (many lines → a dictionary)
4. LeetCode 2161: **Partition by Pivot** (the tidier `for num in nums` version)

💡 Better ways

approaches.md: Word Order, Two Sum (the `seen` dict version), 2149, 2161

🗓️ Then: Sunday review

Claude gives a 10-question quiz on the whole week, logged in quiz-log.md.

✅ **Answers (only after you've predicted!)**

Saturday - S1 `a a ana` · S2 `2 -1` · S3 `3 True False` - S4 `banana`: string methods return a NEW string, so the result must be saved (`s = s.upper()`) - S5 `['a', 'b', 'c']` · S6 `x-y-z` · S7 `a b c` · S8 `0 1 2` - S9 `0 5` / `1 7` / `2 9` · S10 `1, 2, 3` · S11 `AB` · S12 `helo` - S13 **positions loop**, because you need the position: `for i in range(len(s)):` / `if s[i] == "a": print(i)` → `1 3 5` - S14 `ch` is already a **letter**, so `word[ch]` crashes ("string indices must be integers"). Use `if ch == "e":`

Sunday - M1 `['4', '5', '6']` (still text) · M2 `<map object at 0x ... >` (a recipe, not a list) · M3 `[4, 5, 6]` - M4 `['1', '2', '3']` · M5 `1 2 3` · M6 `Harsh ['25', '26']` - D1 `1 None 0` · D2 `a b` (looping a dict gives **keys**) · D3 `a 1` / `b 2` · D4 `[1, 2] 2` - D5 `True False` (`in` checks **keys**, not values) · D6 `{'a': 10, 'b': 2, 'c': 3}` (changing a key keeps its place) - D7 `{'h': 1, 'e': 1, 'l': 2, 'o': 1}` · D8 **KeyError: 'z'** - D9 KeyError on the first `"hi"`: the key doesn't exist yet. Use `counts[word] = counts.get(word, 0) + 1` - D10 `print(" ".join(map(str, nums)))` or `for n in nums: print(n, end=" ")` (or `print(*nums)`)

📘 Day 06: Functions 🧩

↑ contents

A **function** is a named, reusable block of code. You've been *using* them all along (`print()` , `len()` , `max()`). Now you make your own.

1. Define and call

```
def greet():                    # def = define a function
    print("Hello, Gowri!")

greet()                         # call it (and reuse it as often as you like)
```

2. Parameters = input

```
def greet(name):
    print(f"Hello, {name}!")

greet("Optimus")                # Hello, Optimus!
```

3. `return` = output ⭐

```
def square(n):
    return n * n                # hands the answer BACK to the caller

result = square(5)              # 25
print(square(3) + square(4))    # 25
```

⚠️ `print` vs `return` : the #1 trap

```
def add_print(a, b):
    print(a + b)                # only shows it on screen

def add_return(a, b):
    return a + b                # gives the value back so you can use it

x = add_print(2, 3)             # shows 5, but x = None 😱 (just like .sort()!)
y = add_return(2, 3)            # y = 5 ✅
```

4. `return` exits immediately

```
def is_even(n):
    if n % 2 == 0:
        return True             # stops here
    return False                # only reached if the if was False

# shorter: return n % 2 == 0  (the comparison already IS True or False)
```

5. Defaults

```
def power(base, exp=2):
    return base ** exp

power(3)      # 9  (exp defaults to 2)
power(2, 5)   # 32
```

6. Inside stays inside

```
def make_total():
    total = 100
make_total()
print(total)   # ❌ NameError: use return to get values out
```

✍️ Exercise: `day06.py`

Turn your old code into reusable tools 🧰 1. `is_even(n)` → True/False. Test with 4 and 7. 2. `is_palindrome(word)` → True/False, ignoring case (reuse Day 04) 3. `count_vowels(text)` → number of vowels (reuse Day 04) 4. `second_highest(scores)` → handles duplicates (reuse Day 05 / Runner-Up) 5. `average(numbers)` → the average of a list

Call each one and print the result:

```
print(is_palindrome("Racecar"))          # True
print(second_highest([95, 95, 88, 61]))  # 88
```

Bonus: `grade(score)` → `"A"` for 90 or more (90 counts!), `"B"` for 75–89, `"C"` for 50–74, below 50 `"F"`. Use early `return` s, with no `elif` needed!

🔥 HackerRank: Write a function

Return `True` if a year is a **leap year**: - divisible by **4** → leap... - **unless** divisible by **100** → not leap... - **unless** also divisible by **400** → leap again!

Test values: 2000 ✅, 1900 ❌, 2024 ✅, 2023 ❌

Hints: - `year % 4 == 0` checks divisibility - Use early `return` and check the **most specific rule first** (400 → 100 → 4) - Keep HackerRank's `def is_leap(year):` and fill in the logic

📘 Day 07: Dictionaries 📖

↑ contents

A **dictionary** stores **pairs**: a **key** → a **value**. You look things up by **name** instead of by position, just like a real dictionary (word → meaning) or a phone book (name → number).

1. Creating and reading

```
robot = {
    "name": "Optimus",
    "maker": "Tesla",
    "year": 2022,
}
print(robot["name"])        # Optimus  (look up by KEY, not position)
print(len(robot))           # 3 pairs
```

- `{ }` with `key: value` pairs, separated by commas
- Keys are usually strings, and values can be anything (numbers, strings, lists...)

2. Missing keys: `[]` vs `.get()`

```
print(robot["price"])           # ❌ KeyError: 'price'
print(robot.get("price"))       # None, no crash
print(robot.get("price", 0))    # 0, your own default
print("price" in robot)         # False, checks KEYS
```

3. Adding, changing, deleting

```
robot["price"] = 20000      # add a new pair
robot["year"] = 2024        # change an existing value (same syntax!)
del robot["maker"]          # delete a pair
```

Dictionaries **can be changed**, like lists.

4. Looping

```
for key in robot:                   # loops over KEYS
    print(key)

for value in robot.values():        # just the values
    print(value)

for key, value in robot.items():    # BOTH at once ⭐ the most useful
    print(f"{key}: {value}")
```

5. The COUNTING pattern ⭐ (you'll use this constantly)

Think of a tally sheet 🖍️. Read one letter at a time. If the letter isn't on the sheet, write it down with 1. If it's already there, add 1 to its number. The dictionary is the tally sheet: **keys = letters, values = counts**.

```
word = "banana"
counts = {}                         # start empty (ACCUMULATOR!)
for ch in word:
    if ch in counts:
        counts[ch] += 1             # seen before: add 1
    else:
        counts[ch] = 1              # first time: start at 1
print(counts)                       # {'b': 1, 'a': 3, 'n': 2}
```

Watch the sheet change: | Round | `ch` | On sheet? | What happens | `counts` after | |---|---|---|---|---| | 1 | b | no | `counts["b"] = 1` | `{'b': 1}` | | 2 | a | no | `counts["a"] = 1` | `{'b': 1, 'a': 1}` | | 3 | n | no | `counts["n"] = 1` | `{'b': 1, 'a': 1, 'n': 1}` | | 4 | a | **yes** | `counts["a"] += 1` | `{'b': 1, 'a': 2, 'n': 1}` | | 5 | n | **yes** | `counts["n"] += 1` | `{'b': 1, 'a': 2, 'n': 2}` | | 6 | a | **yes** | `counts["a"] += 1` | `{'b': 1, 'a': 3, 'n': 2}` |

Why the `if`? `counts["a"] += 1` means "add 1 to the number next to `a`". If `a` isn't on the sheet yet, there's no number, so it crashes with `KeyError`. The first time, you must write it down with 1.

Shorter, using `.get()` with a default:

```
for ch in word:
    counts[ch] = counts.get(ch, 0) + 1
```

Read it **right side first**: 1. `counts.get(ch, 0)`: "what number is next to this letter? If it's not there, pretend it's **0**" 2. `+ 1`: add one 3. `counts[ch] = ...`: write the new number back

Round 1 (`b`, not there): 0 + 1 = **1**. Round 4 (`a`, already 1): 1 + 1 = **2**. It's the same as the `if/else` version, with the `0` replacing the "not there yet" case.

See it yourself: add `print(ch, "→", counts)` inside the loop and watch the sheet grow.

6. Lists inside dictionaries

```
marks = {
    "Gowri": [85, 92, 78],
    "Optimus": [99, 100, 98],
}
print(marks["Gowri"][1])                        # 92
print(sum(marks["Gowri"]) / len(marks["Gowri"])) # average of Gowri's marks
```

Lists vs dictionaries

	List `[ ]`	Dictionary `{ }`
Find items by	**position** `x[0]`	**key** `x["name"]`
Order matters?	yes	keeps insertion order, but you look things up by key
Best for	a sequence of similar things	**labelled** data, lookups, counting

---

✍️ Exercise: `day07.py`

1. **Robot profile:** make a dictionary for your favourite robot (name, maker, year, height). Print each pair with `.items()`, add a `"price"`, change the `year`, and delete one key.
2. **Phone book:** a dictionary of 3 names → numbers. Ask the user for a name and print the number, or `"Not found"` if it's missing (use `.get()` or `in`, with **no crash**).
3. **Letter counter:** ask for a word and count each letter with the COUNTING pattern. Then print the **most common letter**. *(Hint: loop over `.items()` and keep the BEST SO FAR, like second-highest!)*
4. **Grade book:** `scores = {"Gowri": 92, "Atlas": 74, "ASIMO": 58, "Spot": 81}`. **Reuse your Day 06 `grade()` function** to print each name with their grade: `Gowri: A`.

Bonus 🧠 (LeetCode preview!): given `nums = [2, 7, 11, 15]` and `target = 9`, find the **two positions** whose numbers add up to the target (answer: `0, 1`, because 2 + 7 = 9). *Hint:* as you loop, store each number you've seen in a dictionary `{number: position}`. For each new number, check whether `target - number` is already in the dictionary. This is **LeetCode problem #1, "Two Sum"**, the most famous interview problem!

🔥 HackerRank: Finding the percentage

They give you a dictionary of `name → list of marks`. Print the **average** of one student's marks, with **2 decimal places**.

Hints: - Their starter code already builds the dictionary `student_marks` for you - Look up the student's list, then `sum( ... ) / len( ... )` (Day 05 + Day 06!) - 2 decimals: `print(f"{avg:.2f}")`. The `:.2f` inside the f-string means "2 digits after the point". That's new and useful!